

Transformer-Based Models for

Multi-Modal Document Understanding

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Abstract

We present a novel approach for document understanding that combines vision transformers with large language models to process both textual and visual content simultaneously. Our method achieves state-of-the-art results on multiple benchmarks, improving F1 scores by 8.5% over previous work. We demonstrate the effectiveness of our approach on a diverse set of document types including scientific papers, invoices, and forms. Our model requires 40% fewer parameters than comparable approaches while maintaining superior performance. Code and trained models are available at <https://github.com/example/multimodal-doc>.

Keywords: document understanding, vision transformers, multi-modal learning, natural language processing

1. Introduction

Document understanding is a fundamental task in natural language processing with applications spanning invoice processing, contract analysis, and scientific document indexing. Traditional approaches rely on separate pipelines for text extraction (OCR) and semantic understanding, which leads to error propagation and suboptimal performance.

Recent advances in vision transformers and large language models have opened new possibilities for joint visual-semantic understanding. However, most existing work treats these modalities independently or uses simple fusion mechanisms. In this paper, we propose a tightly coupled multi-modal architecture that learns rich representations by jointly processing document images and text.

1.1 Contributions

- A novel transformer architecture for multi-modal document processing that significantly improves upon existing approaches

- Extensive evaluation on three document understanding benchmarks with detailed ablation studies

- A comprehensive analysis of the trade-off between model size and performance

- Publicly available pre-trained models and code for reproducibility

2. Related Work

2.1 Document Understanding

Early approaches to document analysis relied on rule-based systems and hand-crafted features. With the advent of deep learning, CNN-based methods became dominant for document classification and layout analysis. More recently, transformer-based models have shown superior performance on various document understanding tasks [1, 2, 3].

2.2 Vision Transformers

Vision Transformers (ViT) were introduced by Dosovitskiy et al. [4] and have since become the de facto standard for image understanding tasks. Several improvements have been proposed, including efficient attention mechanisms and better positional encodings [5, 6].

2.3 Multi-Modal Learning

Multi-modal fusion has been explored in various contexts, from video understanding to image captioning. Recent work has focused on designing better fusion mechanisms and pre-training objectives for learning aligned representations [7, 8].

3. Methodology

3.1 Model Architecture

Our model consists of three main components:

Component	Description
Vision Encoder	Vision Transformer processing document images at 224×224 resolution
Text Encoder	BERT-based encoder for textual content extracted via OCR
Fusion Module	Cross-attention layers combining visual and textual representations

3.2 Training Objective

$$L = \lambda_1 \times L_{align} + \lambda_2 \times L_{task} + \lambda_3 \times L_{contrastive}$$

where L_{align} encourages alignment between visual and textual features, L_{task} is the task-specific loss, and $L_{contrastive}$ promotes discrimination between different document types.

4. Experiments

4.1 Datasets

We evaluate our approach on three public benchmarks:

Dataset	Documents	Task
DocVQA	12,767	Visual question answering
RVL-CDIP	400,000	Document classification
FUNSD	149	Form understanding

4.2 Results

Method	DocVQA	RVL-CDIP	FUNSD
LayoutLM v3	81.2%	92.3%	86.5%
DiT-Base	83.5%	94.1%	88.2%
Our Model	89.7%	96.8%	94.7%
Improvement	+8.5%	+4.5%	+8.2%

4.3 Ablation Study

We conduct ablation studies to understand the contribution of each component. Removing the fusion module decreases performance by 4.2% on average, while removing the contrastive loss decreases performance by 2.1%. These results validate our architectural choices.

5. Analysis and Discussion

5.1 Computational Efficiency

Our model achieves superior performance with **40% fewer parameters** compared to DiT-Base (125M vs 208M parameters). Training time is reduced from 48 hours to 28 hours on a single 8-GPU node.

5.2 Error Analysis

The remaining errors are primarily due to:

- Handwritten text in form fields (15% of failures)
- Multi-column layouts with complex spatial relationships (12%)
- Low-resolution document images (8%)
- Documents in languages outside training data (6%)

5.3 Limitations

While our approach shows promising results, it has some limitations. The model requires high-quality document images and struggles with significantly degraded inputs. Additionally, fine-tuning on new document types requires careful hyperparameter tuning due to the complexity of the multi-modal architecture.

6. Conclusion

We have presented a novel multi-modal architecture for document understanding that achieves state-of-the-art results across multiple benchmarks. Our approach tightly couples vision and language modalities through cross-attention mechanisms, leading to significant performance improvements over existing methods. The model is more efficient than comparable approaches and can be readily deployed in real-world applications.

Future work will focus on extending our approach to handle video documents, incorporating layout information more explicitly, and developing better fine-tuning strategies for domain-specific applications.

References

- [1] Li, Y., et al. (2021). LayoutLM: Pre-training of text and layout for document image understanding. In SIGIR.
- [2] Appalaraju, S., et al. (2021). Docformer: End-to-end transformer model for document understanding. In ICCV.
- [3] Xu, Y., et al. (2021). LayoutLMv2: Multi-modal pre-training for visually-rich document understanding. In ACL.
- [4] Dosovitskiy, A., et al. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. In ICLR.
- [5] Shaw, P., et al. (2018). Self-attention with relative position representations. In NAACL.
- [6] Raffel, C., et al. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. JMLR.
- [7] Radford, A., et al. (2021). Learning transferable visual models from natural language supervision. In ICML.
- [8] Zhang, Y., et al. (2021). Align before fuse: Vision and language representation learning with momentum distillation. In NeurIPS.